

August 10, 2022

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: LLHg - NPDES
Pace Project No.: 50321430

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on July 20, 2022. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: LLHg - NPDES

Pace Project No.: 50321430

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268

Illinois Accreditation #: 200074

Indiana Drinking Water Laboratory #: C-49-06

Kansas/TNI Certification #: E-10177

Kentucky UST Agency Interest #: 80226

Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050

Ohio VAP Certified Laboratory #: CL0065

Oklahoma Laboratory #: 9204

Texas Certification #: T104704355

Wisconsin Laboratory #: 999788130

USDA Soil Permit #: P330-19-00257

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SAMPLE SUMMARY

Project: LLHg - NPDES

Pace Project No.: 50321430

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50321430001	Field Blank	Water	07/19/22 13:30	07/20/22 10:08
50321430002	Effluent	Water	07/19/22 13:35	07/20/22 10:08
50321430003	Effluent Duplicate	Water	07/19/22 13:40	07/20/22 10:08
50321430004	Influent	Water	07/19/22 07:00	07/20/22 10:08

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SAMPLE ANALYTE COUNT

Project: LLHg - NPDES

Pace Project No.: 50321430

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50321430001	Field Blank	EPA 1631E	WJW	1	PASI-I
50321430002	Effluent	EPA 1631E	WJW	1	PASI-I
50321430003	Effluent Duplicate	EPA 1631E	WJW	1	PASI-I
50321430004	Influent	EPA 245.1	ILP	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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SUMMARY OF DETECTION

Project: LLHg - NPDES

Pace Project No.: 50321430

Lab Sample ID Method	Client Sample ID Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50321430002	Effluent					
EPA 1631E	Mercury	2.82	ng/L	0.50	08/09/22 09:23	1d
50321430003	Effluent Duplicate					
EPA 1631E	Mercury	2.19	ng/L	0.50	08/09/22 09:01	

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ANALYTICAL RESULTS

Project: LLHg - NPDES

Pace Project No.: 50321430

Sample: Field Blank		Lab ID: 50321430001		Collected: 07/19/22 13:30		Received: 07/20/22 10:08		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	<0.20	ng/L	0.50	0.20	1	08/08/22 14:58	08/09/22 08:02	7439-97-6	

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ANALYTICAL RESULTS

Project: LLHg - NPDES

Pace Project No.: 50321430

Sample: Effluent		Lab ID: 50321430002		Collected: 07/19/22 13:35		Received: 07/20/22 10:08		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level		Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis							
Mercury	2.82	ng/L	0.50	0.20	1	08/08/22 14:58	08/09/22 09:23	7439-97-6	1d

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ANALYTICAL RESULTS

Project: LLHg - NPDES

Pace Project No.: 50321430

Sample: Effluent Duplicate		Lab ID: 50321430003		Collected: 07/19/22 13:40		Received: 07/20/22 10:08		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
1631E Mercury, Low Level Analytical Method: EPA 1631E Preparation Method: EPA 1631E Pace Analytical Services - Indianapolis									
Mercury	2.19	ng/L	0.50	0.20	1	08/08/22 14:58	08/09/22 09:01	7439-97-6	

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ANALYTICAL RESULTS

Project: LLHg - NPDES

Pace Project No.: 50321430

Sample: Influent		Lab ID: 50321430004		Collected: 07/19/22 07:00		Received: 07/20/22 10:08		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
245.1 Mercury Analytical Method: EPA 245.1 Preparation Method: EPA 245.1 Pace Analytical Services - Indianapolis									
Mercury	<0.12	ug/L	0.20	0.12	1	07/25/22 21:41	07/26/22 08:27	7439-97-6	

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QUALITY CONTROL DATA

Project: LLHg - NPDES
Pace Project No.: 50321430

QC Batch:	689973	Analysis Method:	EPA 1631E
QC Batch Method:	EPA 1631E	Analysis Description:	1631E Mercury
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50321430001, 50321430002, 50321430003

METHOD BLANK: 3173011 Matrix: Water
Associated Lab Samples: 50321430001, 50321430002, 50321430003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.20	0.50	0.20	08/09/22 07:55	

METHOD BLANK: 3173012 Matrix: Water
Associated Lab Samples: 50321430001, 50321430002, 50321430003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.20	0.50	0.20	08/09/22 09:16	

METHOD BLANK: 3173013 Matrix: Water
Associated Lab Samples: 50321430001, 50321430002, 50321430003

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ng/L	<0.20	0.50	0.20	08/09/22 10:58	

LABORATORY CONTROL SAMPLE: 3173014

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	ng/L	5	4.85	97	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3173015 3173016

Parameter	Units	50321550006 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ng/L	<0.50	2.5	2.5	3.19	3.05	117	112	71-125	4	24	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3173173 3173174

Parameter	Units	50322662001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ng/L	1.16	2.5	2.5	3.53	3.29	95	85	71-125	7	24	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: LLHg - NPDES

Pace Project No.: 50321430

QC Batch: 687533

QC Batch Method: EPA 245.1

Analysis Method: EPA 245.1

Analysis Description: 245.1 Mercury

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50321430004

METHOD BLANK: 3162881

Matrix: Water

Associated Lab Samples: 50321430004

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Mercury	ug/L	<0.12	0.20	0.12	07/26/22 07:55	

LABORATORY CONTROL SAMPLE: 3162882

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	ug/L	5	5.0	99	85-115	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3162883 3162884

Parameter	Units	50321364002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	ug/L	ND	5	5	5.0	4.9	99	97	70-130	2	20	

MATRIX SPIKE SAMPLE: 3162885

Parameter	Units	50321640001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	ug/L	ND	5	<0.12	0	70-130	M0

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QUALIFIERS

Project: LLHg - NPDES

Pace Project No.: 50321430

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d This sample and the customer-specified duplicate exceeded 20%R RPD. wjw

M0 Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: LLHg - NPDES

Pace Project No.: 50321430

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50321430001	Field Blank	EPA 1631E	689973	EPA 1631E	690236
50321430002	Effluent	EPA 1631E	689973	EPA 1631E	690236
50321430003	Effluent Duplicate	EPA 1631E	689973	EPA 1631E	690236
50321430004	Influent	EPA 245.1	687533	EPA 245.1	687750

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CHAIN-OF-CUSTODY / Analytical Request Document

The Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields must be completed accurately. Consent and acceptance of the Pace Terms and Conditions found at <https://info.pace.com/chain-of-custody>

WO#: 50321430



50321430

Section A

Section B

Required Project Information:

Invoice Information:

Company:	City of Cadillac	Report To:	Cindy Tomaszewski	Attention:
Address:	1121 Plett Rd	Copy To:		Company Name:
	Cadillac, MI 49601			Address:
Email:	ctomaszewski@cadillac-mi.net	Purchase Order #:		Price Quote:
Phone:	231-775-2368	Project Name:	LLHg - NPDES	Price Project Manager:
Requested Due Date:		Project #:		Price Profile #:
				brian.hall@pacelabs.com,

MI

[illegible]



Sample Conditions Upon Receipt Form (SCUR)

Date/Time: <u>7/20 1220</u>	Evaluated By: <u>DAS</u>	WO#: 50321430	
Client: <u>City of Cadillac</u>	PM: <u>BJH</u>	PM: BJH Due Date: 08/03/22 CLIENT: GR-Cadillac	
Lab Notified of Rush or Short Holds: YES ☒ NO ☒			
Project Received Via: FedEx ☒ UPS ☒ Client ☐ Pace Courier ☐ Other: _____	Comments:		
Custody Seal Present and Intact:	YES ☒	NO ☐	NA ☒
Received Sample Information Form (SIF): Drinking Waters Only	YES ☒	NO ☐	NA ☒
Short Hold Present (≤ 48 Hours):	YES ☒	NO ☐	☒
Sample Received in Hold:	YES ☒	NO ☐	☒
Custody Signature Present:	YES ☒	NO ☐	☒
Collector Signature Present:	YES ☒	NO ☐	☒
Sample Collected Today and On Ice:	YES ☒	NO ☐	N/A ☒
IR Gun #: 280 281	Temp. should be 0°C - 6°C (Initial/Corrected)		
Ice Type: WET Bagged / WET Loose BLUE NONE	1. Cooler Temp. Upon Receipt: <u>3.7/3.3</u> °C		
Ice Location: TOP BOTTOM MIDDLE DISPERSED	2. Cooler Temp. Upon Receipt: _____ °C		
Temp Blank Received:	YES ☒	NO ☐	☒
Sample Label Matches COC (ID/Date/Time):	YES ☒	NO ☐	☒
Container Intact:	YES ☒	NO ☐	☒
Correct Container:	YES ☒	NO ☐	☒
Sufficient Volume:	YES ☒	NO ☐	☒
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation pH Strip Lot #: <u>HC 178690</u> Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl	YES ☒	NO ☐	N/A ☒
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ Applies to SVOC 625, PCB/Pest. 608, Total/Amenable Cyanide	YES ☒	NO ☐	N/A ☒
VOA Headspace Acceptable (<6mm):	YES ☒	NO ☐	N/A ☒
Trip Blank Received: HCl MeOH Other: _____	YES ☒	NO ☐	ON HOLD ☒
Comments:	3. Cooler Temp. Upon Receipt: _____ °C		
	4. Cooler Temp. Upon Receipt: _____ °C		
	Non-Conformance Form Required: YES NO		